

Polynomial conditioning of a fixed number of cumulative moments

Proof independently reviewed by root and independent_directions

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A single Legendre perturbation places all its cumulative moments in the polynomial module spanned by $1, H_d, H'_d$, so it cannot condition arbitrarily many independent moment features. A fixed number of separated perturbation bands removes that obstruction.

Theorem 1. *Fix $s \geq 1$, let $r \geq 0$, and put $R = r + s + 1$. Choose positive integers*

$$d_i > 2 \left(R + \sum_{j < i} (d_j + 1) \right) + 1, \quad 1 \leq i \leq s,$$

and put $D = R + \sum_i (d_i + 1)$. For $0 < \eta \leq 1/(2s)$ define the strictly positive density and its cumulative moments

$$f(t) = 1 - \eta \sum_{i=1}^s P_{d_i}(2t - 1), \quad M_l(t) = \int_0^t u^l f(u) du \quad (0 \leq l \leq s).$$

There are constants C_s, A_s such that, for arbitrary $p_, p_0, \dots, p_s \in \mathcal{P}_r$,*

$$\left(\|p_*\|_2^2 + \sum_{l=0}^s \|p_l\|_2^2 \right)^{1/2} \leq C_s (D + 1)^{A_s} \eta^{-2s} \left\| p_* + \sum_{l=0}^s p_l M_l \right\|_2.$$

All norms are for Lebesgue measure on $[0, 1]$. Minimal degree choices have $D = O_s(r + 1)$. The same bound, up to fixed constants, holds in any measure whose density is bounded above and below by fixed positive constants. It also holds empirically on a real quadrature exact to at least twice the degree of the displayed combination.

Proof. Write $w = t(1 - t)$, $H_i(t) = \int_0^t P_{d_i}(2u - 1) du$, and $\lambda_i = d_i(d_i + 1)$. For $l \geq 1$, integration by parts gives

$$M_l = \frac{t^{l+1}}{l+1} - \eta \sum_i \left[t^l H_i - l \int_0^t u^{l-1} H_i(u) du \right].$$

For $l = 0$, $M_0 = t - \eta \sum_i H_i$. The exact primitive lemma in `ordered_prefix_legendre_tools.tex` uses $\mathcal{L}_i C = (wC)'' + \lambda_i C$ and gives

$$\int_0^t u^{l-1} H_i(u) du = A_{il}(t) H_i + B_{il}(t) H'_i, \quad B_{il} = -w \mathcal{L}_i^{-1} t^{l-1}.$$

Here $\deg A_{il} \leq l$ and $\deg B_{il} \leq l + 1$; their coefficients and norms are bounded by a fixed power of D for fixed s . Consequently the combination $F = p_* + \sum_l p_l M_l$ has the form

$$F = P + \eta \sum_i (H_i A_i + H'_i B_i), \quad \deg P, \deg A_i, \deg B_i \leq R,$$

where, crucially,

$$B_i = -w \sum_{l=1}^s l p_l \mathcal{L}_i^{-1} t^{l-1}. \quad (1)$$

The trinary-band conditioning theorem from the same note yields

$$\|P\|_2 + \sum_i (\|A_i\|_2 + \|B_i\|_2) \leq C_s(D+1)^{6s} \eta^{-s} \|F\|_2.$$

We show that the B_i recover $p_1, \dots, p_s$ polynomially. Let $J_0, \dots, J_{s-1}$ be the shifted Jacobi polynomials for weight w , with any fixed rational normalization. They satisfy

$$\mathcal{L}_i J_j = (\lambda_i - \mu_j) J_j, \quad \mu_j = (j+1)(j+2).$$

Expand $t^{l-1} = \sum_{j=0}^{l-1} a_{jl} J_j$ and set $c_j = \sum_{l=j+1}^s l a_{jl} p_l$. This is a fixed invertible triangular transformation from $(p_1, \dots, p_s)$ to $(c_0, \dots, c_{s-1})$. Equation (1) becomes

$$-B_i = \sum_{j=0}^{s-1} \frac{w J_j c_j}{\lambda_i - \mu_j}.$$

The scalar Cauchy matrix $C_{ij} = 1/(\lambda_i - \mu_j)$ is invertible. Its determinant has absolute value

$$\frac{\prod_{i < i'} |\lambda_{i'} - \lambda_i| \prod_{j < j'} |\mu_{j'} - \mu_j|}{\prod_{i,j} |\lambda_i - \mu_j|}.$$

All differences are nonzero integers, and each denominator factor is at most $C_s(D+1)^2$. Thus the determinant is bounded below by an inverse polynomial in D , while all entries and cofactors are bounded by constants depending only on s . The inverse matrix has polynomial norm. It follows that each $\|w J_j c_j\|_2$ is at most $\text{poly}_s(D) \eta^{-s} \|F\|_2$.

Multiplication by any fixed nonzero polynomial Q of degree at most $s+1$ has a polynomial lower bound on $\mathcal{P}_r$:

$$\|Qp\|_2 \geq c_Q(r+1)^{-2(s+1)} \|p\|_2.$$

To see this, remove intervals of length $c_Q(r+1)^{-2}$ around its finitely many real roots in $[0, 1]$. The polynomial evaluation bound retains at least half the squared norm of p on the remainder, and factorization of Q gives the stated pointwise lower bound there. Apply this to the fixed polynomials $w J_j$. We recover all c_j , and therefore all p_l , $l \geq 1$, with polynomial loss.

Since f is a probability density between $1/2$ and $3/2$, $0 \leq M_l \leq 1$. Subtracting $\sum_{l=1}^s p_l M_l$ from F therefore leaves $p_* + p_0 M_0$ with polynomially bounded norm. Its trinary representation has constant coefficient $p_* + t p_0$ and each H_i coefficient $-p_0$. One more application of the same conditioning theorem bounds p_0 and then p_* , at the cost of another η^{-s} . This proves the displayed result. The weighted and exact-quadrature statements follow immediately by norm comparison and exactness for the squared polynomial. $\square$

Moment compatibility against a polynomial density. Suppose an additional density v has degree e , and take the first band degree larger than $e + m + s + 1$. For $0 \leq a \leq m$ and $0 \leq l \leq s$, Fubini and Legendre orthogonality give

$$\int_0^1 t^a M_l(t) v(t) dt = \frac{1}{l+1} \int_0^1 t^{a+l+1} v(t) dt.$$

Indeed the perturbation integral is against $P_{d_i}(2u-1)$ times $u^l \int_u^1 t^a v(t) dt$, whose degree is at most $e + a + l + 1 < d_i$. This provides a polynomially conditioned real baseline for rational cumulative-moment ranks. Their constant targets depend only on the first $s+1$ moments of v .

Arithmetic limitation for iterated constructions. The fixed- s box-smoothing theorem additionally preserving its first s conditional moments needs targets with $a \leq s$ in the preceding identity, hence input moments through degree $2s + 1$. Furthermore, polynomial denominator bounds for separate local constructions do not imply a polynomial *common* denominator across polynomially many cells. Synchronizing those denominators is a separate requirement; no construction with arbitrarily many fixed small dice is inferred here.